



BAGAS NOUVAL LIANSYAH

Graduate, D-IV in Telecommunications Electronics Engineering

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PROFILE

Applied Bachelor's (D-IV) graduate in Telecommunications Electronics Engineering from Politeknik Caltex Riau, graduated Cum Laude (GPA 3.54/4.00). Combines technical and managerial expertise with hands-on experience in telecommunications infrastructure maintenance, smart system development, and *Green Energy Solution* design. Proven in executing projects *end-to-end*, from technical simulation and budget planning (RAB) to field operational *troubleshooting*. As a leader experienced in overseeing hundreds of students in organizations, I have become meticulous, disciplined, responsible, adaptive, strategic in communication, and sharp in problem-solving to deliver real contributions to the company.

EDUCATION

Politeknik Caltex Riau	Pekanbaru, Indonesia.
D-IV in Telecommunications Electronics Engineering	2022 - 2026

- GPA 3.54/4.00 (Cum Laude)
- Final Project Title: IoT-Based Notification System for Water Dispensers at Politeknik Caltex Riau

SMA Negeri 14 Pekanbaru	Pekanbaru, Indonesia.
Science (Natural Sciences)	2019 - 2022

SKILLS

- **Electronics & Smart Systems:** Internet of Things (IoT) and Embedded Systems, Programmable Logic Control (PLC), Analog and Digital Electronic Circuits, Electrical Measurement and Instrumentation, PV System Design, and Electromechanical Workshop practice.
- **Telecommunications Technical & Engineering:** Optical Fiber Communication Engineering with Fiber Optic Installation practicum, Radio Frequency (RF) Engineering, Antennas and Propagation, Wireless, Satellite, and Terrestrial Communication Systems, Computer Networks (Cisco Networking), Electromagnetic Compatibility, and comprehensive understanding of Telecommunications Standards and Regulations.
- **Software & Programming:** Microsoft Office (including Visio), C/C++ (Arduino IDE), Python, MATLAB, OptiSystem, CST Studio Suite, HelioScope, Cisco Packet Tracer, Wireshark, Proteus, Eagle CAD, EasyEDA, Tinkercad, PLC Software, IoT Software.
- **Interpersonal Skills (Soft Skills):** Strategic Leadership, Crisis Management & Problem Solving, Teamwork, Effective Communication, High Adaptability, Time Management, Attention to Detail, and *Enabling Skills*.
- **Languages:** Indonesian (Native), English (*Intermediate*)

PROJECTS

- **IoT-Based Notification System for Water Dispensers:** designed a dispenser gallon monitoring & notification system using ESP32, Load Cell, HX711, Blynk, Telegram Bot, and Google Sheets, achieving an average sensor accuracy of 99.82%–100% (error 0–0.19%) and a 100% notification success rate.
- **Fiber Optic Transport Network Design:** Simulated a 10 km fiber optic transmission architecture using OptiSystem software. Performed computational analysis of *Link Power Budget* and *Rise Time Budget* to validate network reliability and minimize *attenuation loss*.
- **Microstrip 2×1 Array Antenna Design & Simulation:** Engineered a *microstrip array* antenna at a frequency of 2.4 GHz using *software* 3D CST Studio. Optimized dimensions and *feeding* techniques to achieve ideal radiation performance with a *Gain* of 5.315 dB, a *VSWR* ratio of 1.11, and an impedance of 50 ohms.
- **Line Follower Robot:** Built an autonomous robot navigation system driven purely by analog circuitry without a microcontroller. Utilized an *Operational Amplifier* (Op-Amp) as a comparator for sensor readings (*photodiode/infrared*) to control the DC motor actuator, achieving 100% accuracy in following sharply maneuvering tracks.

WORK EXPERIENCE

PLN Icon Plus

Pekanbaru, Indonesia.

Internship Staff, Marketing & Sales Division (Customer Solution Sub-Division)

Oct 2025 - Feb 2026

- Managed the end-to-end provision cycle of Rooftop Solar Power Plant solutions (*PV Rooftop*) for PLN Icon Plus customers, starting from *assessment* of electricity needs, *field survey*, through to solution finalization and permit processing.
- Conducted detailed field inspections using *drone* to verify roof structure, area size, panel orientation, and potential *shading* to ensure the system could be synchronized with the PLN grid.
- Designed the topology and technical design of the PV Rooftop system using HelioScope software, formulating a system configuration with a capacity of 161.82 kWp (135 kWac) comprising 261 solar panels (620 Wp) and 3 *inverter* units (45 kW).
- Prepared transparent indicative cost budget plans (RAB) and designed business payment scheme options (*Direct Investment* and *Leasing* for 5-15 years) tailored to customer needs in the *Commercial & Industry* (C&I) segment.

PT Sidmunth Jaya Investema

Pekanbaru, Indonesia.

Internship Staff, Technical Experience (Telecommunications Sector)

Feb 2025 - Jun 2025

- Carried out *preventive* and *corrective maintenance* on dozens of fiber optic infrastructure points (OLT, ODC, FAT) of PLN ICON+ in the Pekanbaru area to prevent and resolve customer connectivity disruptions.
- Analyzed and resolved optical fiber cable break issues using the *tracing* method, and performed precise *core* splicing using a *Fusion Splicer* both in the field and at the *Joint Box*.
- Operated industry-standard telecommunications measuring instruments (OTDR, OPM, VFL, etc.) to ensure normal signal attenuation values.
- Coordinated in real time with the *Network Operation Center* (NOC) team to verify system recovery after repairs, and documented daily repair reports in accordance with occupational health and safety (K3) standards.

ORGANIZATIONAL EXPERIENCE

Association of Electronics Telecommunication (AET) Politeknik Caltex Riau

Pekanbaru, Indonesia.

Head of Commission B (Legislative Body)

2025 - 2026

- Rigorously audited and validated the financial accountability reports (LPJ) of the executive body.
- Formulated strategic improvement recommendations to the Study Program based on evaluation and research on student aspirations.

Junior Prime Minister / Head of Student Association (Executive Body)

2024 - 2025

- Led, coordinated, and held full responsibility for all operations, activities, and strategic policies of the organization at the study program level.
- Created and executed innovative flagship work programs based on *data-driven* approaches, benefiting 4 cohorts of students.
- Managed budget flow (RAB) and successfully secured additional operational funding through strategic *sponsorship* partnerships.

Junior Prime Minister (Junior Executive Body)

2023 - 2024

- Controlled and ran the daily operations of the executive body and served as a strategic communication facilitator between newer and senior generations of students.
- Carried out the leadership succession phase (*succession planning*) to accelerate managerial capabilities in preparation for becoming the next top leader.

CERTIFICATES

- **Cisco Networking Academy** — CCNA: Introduction to Networks 2026
- **BNSP Certification** — Intermediate Technician of Computer Networks 2026 - 2029
- **Intensive TOEIC Certificate** 2026